INCH-POUND

GGG-B-500D
27 March 2015
SUPERSEDING
GGG-B-500C
11 October 1983

FEDERAL SPECIFICATION

BLOCK, TACKLE (WIRE ROPE)

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers single-sheave snatch and single- and multiple-sheave ordinary wire rope tackle blocks for regular and heavy duty use.

1.2 Classification.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity indicated by the title of the specification, but cover only those most generally used by the Federal Government.

1.2 Classification. Tackle blocks shall be of the following types, grades, and styles, as specified (see 6.2):

Type I - Snatch blocks single sheave, safety locking.
Type II - Ordinary blocks.
Grade A - Regular duty.
Grade B - Heavy duty.
Style 1 - Single sheave .
Style 2 - Double sheave.
Style 3 - Triple sheave.

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AMSC/NA FSC 3940

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2. APPLICABLE DOCUMENTS

2.1 Government documents .

2.1.1 Specifications, standards. and handbooks. The following specifications , standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those Listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)

Federal Standards:

FED-STD-H28 - Screw Thread Standards for Federal Services

FED-STD-123 - Marking for shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, Commercial Item Descriptions, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas C1ty, MJ; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents as the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of US Military Property

MIL-STD-889 - Dissimilar Metals

(Copies of these documents are available online at http://quicksearch.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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2.2 Other publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not 1isted in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture Surface Roughness, Waviness and lay.

(Copies of these documents are available from http://www.ansi.org or the American National Standards Institute, ATTN: Sales Dept., 25 West 43rd Street, 4th Floor, New York, NY 10036).

ASTM International

ASTM A36/A36M - Structural Steel

ASTM A148/A148M - High Strength Steel Castings for Structural Purposes

ASTM A153/A153M - Zinc Coating (Hot Dip) on Iron and Steel Hardware

ASTM A675/A675M - Steel Bars, Carbon, Hot Wrought, Special Quality, Mechanical Properties

ASTM B438 - Bronze-Base Powder Metallurgy (PM) Bearings (Oil-Impregnated) Bearings

ASTM D3951 - Commercial Packaging, Standard Practice for.

(Copies of these documents are available from http://www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

Society for Automotive Engineers (SAE)

AS15003 - Fittings, Lubrication (Hydraulics) Surface Check, 1/8 Pipe Threads, Steel, Type III.

(Copies of this document are available online from http://www.sae.org or the Society for Automotive Engineers, 400 Commonwealth Drive, Warrandale, PA 15096-001)

American Society for Quality (ASQ):

ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes..

(Copies of this document are available online from http://asq.org or from the ASQ Distribution Center, 5131 S. Third Street, Milwaukee, Wisconsin 53207-6028.)

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3. REQUIREMENTS

3.1 Description. The blocks shall consist of a shell, one or more sheaves and straps combined with all necessary interconnecting fittings to form a complete unit assembly.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practical, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.3.2 Material deterioration and control. The block shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.2.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.2.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components , and shall make information available, upon request to the contracting officer or designated representative.

3.4 Screw threads All screw threads shall conform to FED-STD-H28.

3.5 Strength

3.5.1 Proof load. When subjected to a proof load of twice the safe working load (SUL) specified in Tables I and II, the block shall show no evidence of deformation, distortion, cracks or permanent set of any component.

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Table I. Type I, snatch blocks

Sheave size OD	Rope dia.	Minimum diameter at depth of sheave groove	SWL
Inches	Inches	Inches	(Lbs.)
6	3/8 to 1/2	4-1/2	4,000
8	1/2 to 5/8	6-1/8	6,000
10	5/8 to 3/4	7-3/4	8,000
12	3/4 to 7/8	9-3/8	10,000
14	7/8 to 1	11	12,000
16	1	13	15,000

Table II. Type 11, regular blocks

Sheave size OD	Grade A				Grade B			
	Rope dia.	Style			Rope dia.	Style		
		1 SWL	2 SWL	3 SWL		1 SWL	2 SWL	3 SWL
Inches	Inch	Lbs.	Lbs.	Lbs.	Inch	Lbs.	Lbs.	Lbs.
6	3/8 to 1/2	3,000	4,000	5,000	1/2 to 5/8	4,000	4,500	6,000
8	1/2 to 5/8	4,000	5,000	6,000	5/8	5,000	7,000	9,000
10	5/8	5,000	6,000	7,000	5/8 to 3/4	7,000	9,000	10,000
12	5/8 to 3/4	6,000	9,000	11,000	3/4 to 7/8	9,000	10,000	12,000
14	3/6 to 7/8	9,000	11,000	13,000	7/8 to 1	10,000	12,000	16,000
16		--	--	--	1	13,000	16,000	20,000

Note: For minimum diameter at depth of sheave groove , refer to type I of table I.

3.5.2 Ultimate strength. When tested for ultimate strength, the fittings, hook, or shackle shall not fail at loads less than 4 times the SWL specified in tables I and II. The block shall then be further tested to destruction but shall not fail at loads less than 5 times the SWL specified in tables I and II.

3.6 Fastening devices. Bolts, nuts, rivets, shell pipe sleeves, cotter pins, and other items used as Fasteners or spreaders , shall be made of commercial quality steel commonly used for this purpose. Such parts shall not be swaged, peened, stacked , or otherwise permanently deformed.

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3.7 Shell plates. Shell plates shall be steel conforming to ASTM A36/A36M. The shell plates shall fully shield the sheave and shall have smooth edges. Plates shall be held in position with bolts, nuts, and pins to prevent distortion or spreading. Each block shall have the shell plates held together at the bottom with a minimum of three bolts with nuts, and tubing spacers; the center bolt shall extend through each strap piece. Type I shell plates shall have an impression or indentation guard pressed into its side extending through a minimum of 180 degrees around the top of the shell in a manner to prevent the tire rope from being jammed between the shell and the sheave. Type II shell plates shall be diamond or oval pattern as specified (see 6.2).

3.8 Straps. Straps shall be steel conforming to ASTM A675/A675M. Straps shall be two-piece construction and shall extend the full length of each shell plate. Straps shall be securely welded, riveted, or bolted to the shell plates in such a manner as to securely hold the shell plates in position and to add rigidity to the block assembly. The straps shall provide bearing support for the center pin and connections to transmit loads from the sheave to the fittings.

3.9 Sheaves. Sheaves shall be cast steel conforming to ASTM A148/A148M, grade 80-40, and shall have a minimum hardness of Rockwell 34. Sheave groove depth shall be as specified in table I and shall be designed to prevent cutting action on the rope. The rim and tread section shall be grooved to fit the contour of the reps under load conditions. The sheaves shall be concentric with the bore, and shall be designed so that only the hubs bear against the shells. Side play shall not exceed .125 inch. Sheaves shall rotate freely when rotated by hand.

3.10 Center pin. Center pin shall be cold drawn medium carbon steel having a carbon content of .25 to .50 percent. The center pin shall be locked in position to prevent rotation. The center pin shall be fitted with key head and hex nut with cotter key. When specified (see 6 .2), grease channels shall be drilled through the pin and a lubricating fitting shall be recessed into the head of the pin to provide lubrication of the bearing surfaces. Lubrication fittings shall conform to SAE AS15003 Surface roughness of center pin bearing surfaces shall not exceed 63 micro inches determined in *accordance* with ANSI B46. 1.

3.11 Fittings. Fittings shall be forged medium carbon steel having a carbon content of .25 to .50 percent, and shall swivel freely through 360°.

3.12 Galvanizing. All metal surfaces of blocks except bearing surfaces shall be hot dipped galvanized in accordance with ASTM A153/A153M.

3. 13 Identification marking. The blocks shall be identified in accordance with MIL-STO-130. Marking shall be permanently and legibly stamped on a shell plate indicating the SWL, size of wire rope to be used with the block, and the manufacturer's code.

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3.14 Type I blocks. Type I blocks shall be as shown in figure 1 (see 6.4); and shall conform to the requirements specified in. table I. The locking device shall be locked into position when the fittings and crosshead are rotated 90º from its normal load position.

3.14.1 Bearings. Bearings shall conform to ASTM B438, material code CT-1000-K26.

3.14.2 Fittings. Fittings shall be as specified (see 6.2), and shall be as shown on figures 1 and 2 (see 6.4).

3.15 Type 11 blocks. Type II blocks shall be as shown in figure 3 (see 6.4), and shall conform to the requirements specified in table II. Blocks shall be supplied with a becket. Fittings shall be as specified (see 6.2), and shall be as shown on figures 3 and 4.

3.15.1 Grade A, regular duty. Bearings for grade A blocks shall conform to ASTM B438, material code CT-1000-K26.

3.15.2 Grade B, heavy duty. Bearings for grade B blocks shall be pressure lubricated, phosphor-bronze. The bearing shall be pressed into the hub of the sheave with sufficient tightness to prevent slippage under any load. Blocks shall be delivered with a high grade of commercial lubricant .

3.15.3 Style 1, single sheave. Style 1 blocks shall have a single sheave.

3.15.4 Style 2, double sheave. Style 2 blocks shall have two sheaves

3.15.5 Style 3, triple sheave. Style 3 blocks shall have three sheaves.

3.16 Workmanship. The blocks shall be free from any characteristics or defects that may render them unsuitable or inefficient for intended purposes All parts of the assemblies shall be clean, free of dirt, sand, fins, pits, flux, and extraneous material . (Welding shall not be resorted to as a repair measure.)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows :

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of preparation for delivery (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article block shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article block shall be tested as specified in 4.5.2.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 For examination and proof load test. Sampling for examination and proof load test shall be in accordance with ASQ Z1.4, Inspection level II.

4.4.1.2 For ultimate test. One block assembly shall be selected at random from each lot

4.4.2 Examination. Samples selected in accordance with 4.4.1.1 shall be examined as specified in 4.5.1.

4.4.3 Tests.

4.4.3.1 Proof load. Samples selected in accordance with 4.4. 1.1 shall be tested as specified in 4.5.2.1.

4.4.3.2 Ultimate load. Samples selected in accordance with 4.4. 1.2 shall be tested as specified in 4.5.2.2. Failure of the test shall be cause for rejection of the lot.

4.5 Inspection procedure.

4 .5.1 Examination. The blocks shall be examined as specified herein for the following defects:

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Major

101. Dimensions not in conformance with table I.
102. Material not as specified.
103. Materials are not resistant to corrosion of deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
104. Dissimilar metals as defined in MIL-sTD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes or treatments.
106. Components missing or improperly assembled.
107. Shell plate rope guard not as specified.
108. Shell plates do not fully shield the sheaves.
109. Shell plates have insufficient number of bolt and tubing spacers.
110. Shell plates twist out of line under load conditions.
111. Cast steel for the sheaves not as specified, contains blowholes, hard spots , shrinkage , or cracks.
112. Sheaves fail to turn freely by hand.
113. Sheaves side play exceeds 1/8 inch.
114. Sheave not designed to prevent cutting action on the rope.
115. Sheave score does not accommodate the rope.
116. Sheave rim not concentric with bore.
117. Bearings not tightly pressed into hub of sheaves. (Type II, grade B only.)
118. Center pin grease fitting, when specified , missing.
119. Center pin grease grooves omitted.
120. Center pin surface roughness exceeds 63 microinches.
121. Center pin rotates in shell.
122. Fittings fail to swivel freely through 360º.
123. Locking device does not function in manner specified, type I only.

Minor

201. Fastening devices peened, stacked, or deformed.
202. Identification marking missing, incomplete , or illegible.
203. Workmanship is not as specified.

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4.5.2 Tests.

4.5.2.1 Proof load. Blocks shall be tested in a testing machine equipped with a dial gage which will indicate the test load. Pass wire rope or round steel bar loops around the sheave or sheaves and attach to the fixed head of the testing machine. Attach blocks and fittings to the moving head of the machine. Steadily apply load to the block until twice the applicable SWL specified in tables I and II is reached. After the test of the block has been completed, the block shall be disassembled and all component parts examined and measured. The divider method shall be used to measure for deformation and the magnetic partial inspection method shall be used to determine presence of cracks in all load carrying parts of the block. Nonconformance with 3.5.1 shall constitute failure of this test.

4.5.2.2 Ultimate load.

4.5. 2.2.1 Fittings. Install the block in a testing machine as specified in 4.5.2.1. Steadily apply load to the block assembly until failure of the fitting occurs. Failure of the fittings at any load less than 4 times the SWL specified in tables I and II, shall constitute failure of this test.

4.5.2.2.2 Block. Attach the block without fittings to the moving and fixed heads of the t-g machine by any suitable means Steadily apply load to the block until failure of the block occurs. Failure of the block at any load less than 5 times the SWL specified in tables I and II, shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack. The preservation, packaging, packing, palletization and marking shall be examined to determine compliance with the requirements of section 5 herein.

5. PACKAGING

5.1 Packaging requirements. Packaging requirements shall be in accordance with ASTM D 3951.

5.2 Marking for civil agencies. Marking for all containers shall be in accordance with FED-STD-123.

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6. NOTES:

6.1 Intended use. Blocks are used to change direction of a running rope or to gain a mechanical advantage which includes hoisting and hauling operations. They are general purpose blocks recognized as a standard product by the tackle industry.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. When a first article is required for inspection and approval and number of units required (see 3.2).

d. Whether shell pattern shall be oval or diamond shaped (see 3.7).

e. Whether grease groove and fitting shall be provided in the center pin (see 3.10).

f. Degree of packaging and packing required (see 5.1 and 5.2).

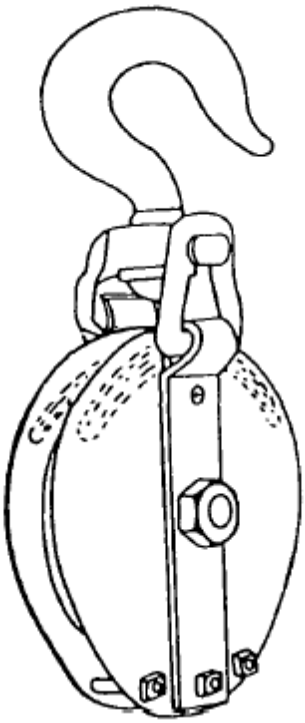
6.3 First article. When a first article inspection is required , the item will be tested and should be a preproduction model. The first article should consist of one or q ore units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations , tests, and approval of the documents first article.

6.4 Figures. The figures show types of blocks which have been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other blocks which conform to this specification.

6.5 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement (QSTAG 396). When amendment, revision, or cancellation of this specification is proposed which affects or violates the international agreement concerned, the preparing activity will inform GSA so that appropriate reconciliation action may be taken through international standardization channels.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TYPE I.

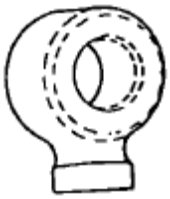
Figure 1. Block, tackle, snatch type I.

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Oblong Swivel Eye



Drilled Swivel Eye



Oblong Swivel Eye And Shackle

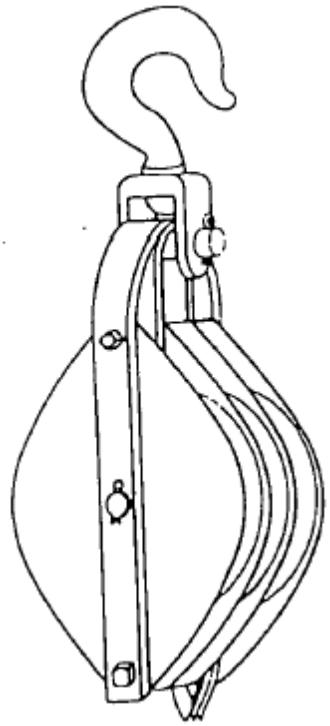


Drilled Swivel Eye And Shackle

Figure 2. Fittings for type I.

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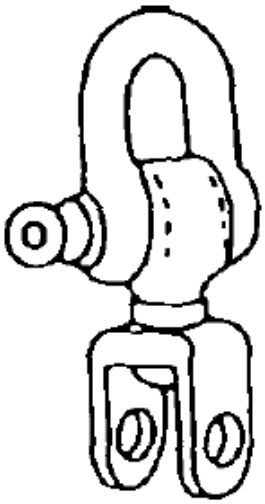


Type II

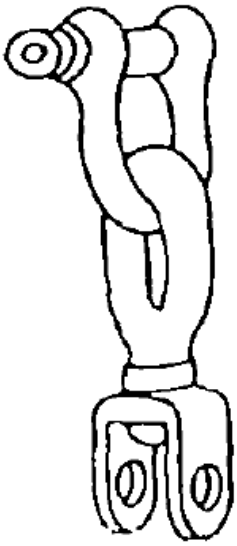
Figure 3. Blocks, tackle ordinary and regular type II, grades A and B.

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Drilled Swivel
Eye And
Shackle



Oblong Swivel
Eye And
Shackle



Drilled Swivel
Eye



Oblong Swivel
Eye

Figure 2. Fittings for type II, grades A and B blocks.

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